

SECTION 1: EXECUTIVE SUMMARY

In today's competitive sports landscape, raw performance data has become a baseline requirement. For PlayerData, the next frontier of growth for their platform lies within, contextual intelligence, or the ability to transform individual athlete metrics into comparative benchmarks. Our Applied Sports Event Research class has partnered with PlayerData to build a scalable framework for these metrics. By orchestrating a cross-functional pipeline, we have delivered a solution consisting of high-fidelity data synthesis, statistical athlete norms for women's soccer, and a multi-platform visualization suite. This report documents the governance and technical execution of this framework, providing a blueprint for immediate in-app enhancement.

SECTION 2: PROBLEM FRAMING & CONSULTING SCOPE

2.1 The Market Opportunity & Problem Statement

PlayerData's value proposition is built upon delivering actionable insights. However, a "Context Gap" exists where users can see *what* an athlete did, but lack the benchmarks to understand *how* that performance ranks against peers. The IU Indy team was tasked with evaluating current metrics to bridge this gap, developing robust comparative statistics, and improving in-app analytical features to support data-driven decision-making for organizations as a whole.

2.2 Project Management & Governance

To address this, the PM team established a **Governance Prototype** to ensure technical accuracy and velocity. We managed the "Control Room" via a centralized Google Drive for documentation and a GitHub Kanban system for task orchestration. This ensured that strategic check-ins on **March 4th** and **April 8th** directly informed the technical builds of the Engineering, Science, and Analytics teams.

SECTION 3: DATA ARCHITECTURE & INFRASTRUCTURE (Data Engineering)

The framework rests on **7,868 sessions** across **281 unique athletes**. A 70-minute **active_minutes** floor was implemented to remove low-intensity sessions, a 35 **max_speed_kph** cap was implemented to exclude erroneous measurements and **Session_load** was identified as the primary composite indicator of training intensity.

- **Synthesis & Stress-Testing:** The team generated two synthetic datasets (10,000 sessions each) for Men's Div II and Women's Div III, validating that performance metrics shift meaningfully across different populations.
- **Conversational AI Agent:** The DE team prototyped a pipeline allowing an LLM to return contextual results in natural language (e.g., *"Jordan's session load was in the 78th percentile for his team"*).

SECTION 4: PERFORMANCE MODELING & ATHLETE NORMING (Data Science)

4.1 Methodology & Feedback Alignment

Following feedback sessions with PlayerData, the DS team narrowed its focus to four high-impact metrics: **Total Distance**, **Max Speed**, **Session Load**, and **Sprint Events**. The team applied weights based on active minutes, filtered for elite training thresholds (>70 min) and eliminated sessions deemed as including improper measurement (max_speed_kph > 35). To ensure the data was "readout-ready" for coaches, the team aggregated athlete sessions by age group (18–21) to establish normative averages and standard deviations.

4.2 Normative Insights & Detailed Results

The following summary statistics tables allow coaches and athletes to view their performance across various metrics and compare them to their respective peers. The average indicates the expected athlete performance for the specific metrics within that age group. The standard deviation indicates the spread of the metric within each age group. The 90th percentile indicates athletes that perform at a higher level in those metrics compared to the rest while the 10th percentile indicates athlete sessions that underperform compared to other sessions.

Total Distance (meters)

Athlete Relative Age	Session Count	Average	Standard Deviation	90th Percentile	10th Percentile	Max.	Min.
18	1472	3853.86	2422.99	7018.37	454.97	11982.59	113.00
19	306	5296.34	1871.28	7706.43	2958.24	9200.14	134.60
20	1086	4063.22	1943.84	6761.51	1580.41	10785.97	100.02
21	334	3641.52	1603.80	5390.36	1683.18	9777.45	106.04

Session Load

Athlete Relative Age	Session Count	Average	Standard Deviation	90th Percentile	10th Percentile	Max.	Min.
18	1472	544.76	412.95	1093.27	37.17	2313.27	4.30
19	306	701.18	301.25	1010.61	369.86	2685.77	7.61
20	1086	542.82	315.60	955.34	189.57	2772.07	4.93
21	334	543.40	318.65	988.64	210.51	1986.89	6.52

- **Total Distance & Session Load:** A significant performance peak was identified in the **19-year-old cohort**, with an average distance of **5,296m** and a session load of **701.18**. This suggests a "sophomore-year" intensity peak. Conversely, 18-year-olds showed the highest standard deviation in distance (2422.99), indicating high variability in freshman adaptation to collegiate training.

Max Speed (kilometers per hour)

Athlete Relative Age	Session Count	Average	Standard Deviation	90th Percentile	10th Percentile	Max.	Min.
18	1472	20.30	5.86	26.28	9.09	31.65	4.18
19	306	21.89	3.91	25.96	17.83	28.88	5.27
20	1086	21.55	3.95	25.51	17.34	30.73	4.54
21	334	21.76	4.13	26.63	16.21	33.89	5.78

- **Max Speed Consistency:** Max speed remained remarkably stable across all groups, with 90th percentile values consistently hitting **~26 kph**. However, the 10th percentile for 18-year-olds was nearly half that of other groups, suggesting a wider gap in sprint-readiness for incoming athletes.

Sprint Events

Athlete Relative Age	Session Count	Average	Standard Deviation	90th Percentile	10th Percentile	Max.	Min.
18	1472	1.28	2.29	4	0	16	0
19	306	1.67	2.44	5	0	15	0
20	1086	1.29	2.39	4	0	20	0
21	334	2.12	3.70	7	0	26	0

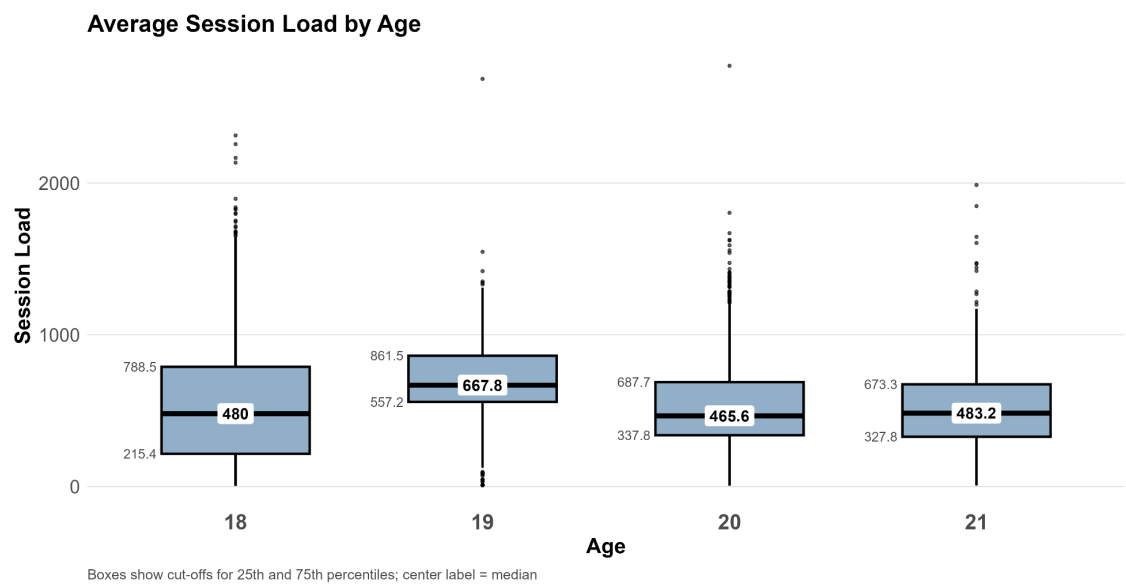
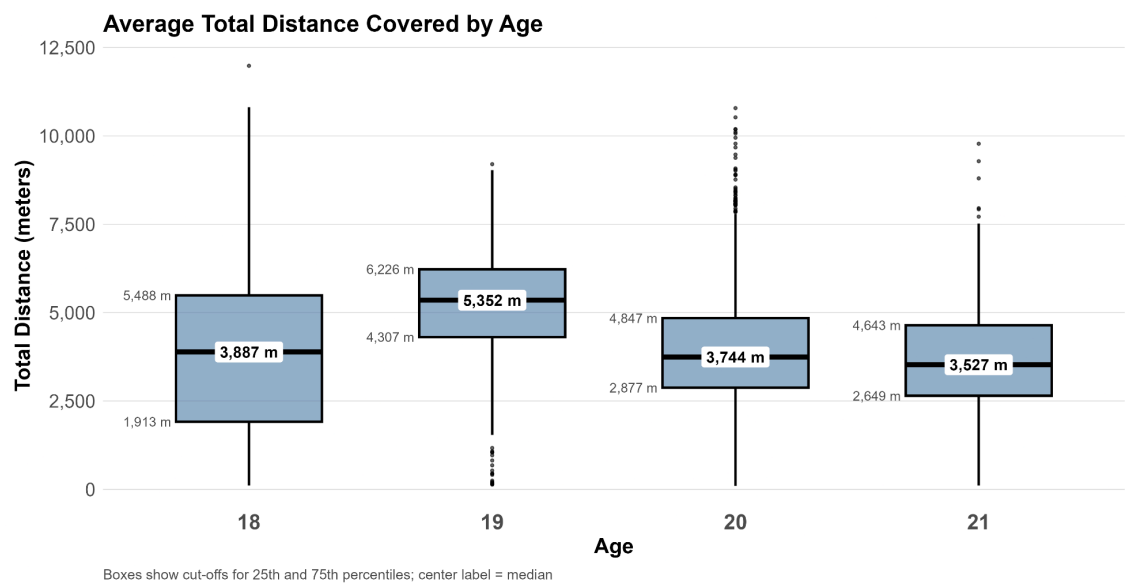
- **Sprint Events:** 21-year-olds averaged 2.12 sprint events—the highest of any group—and had the highest standard deviation. This identifies a specific subset of "higher-explosivity" seniors that the platform can now isolate.
- **Data Imbalance Note:** The team observed that 18- and 20-year-olds contributed over 1,000 sessions each, while 19- and 21-year-olds contributed roughly 300. While this imbalance is a cause for concern in absolute generalization, the prototype findings remain statistically significant for this collegiate sample.

SECTION 5: INTERACTIVE VISUALIZATION & PROTOTYPE DESIGN (Business Analytics)

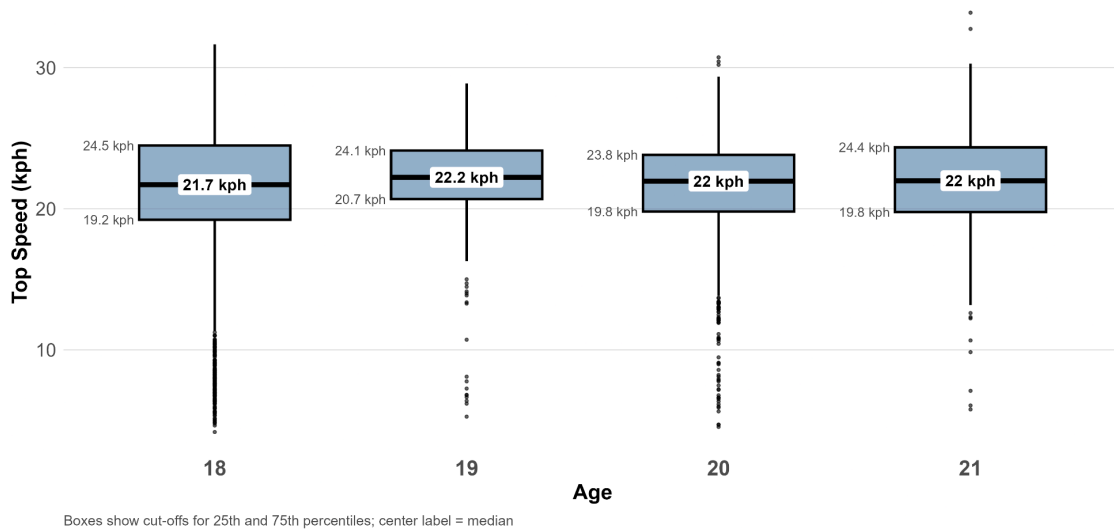
5.1 Design Philosophy & Platform Exploration

The BA team focused on translating DS norms into a "Coach-First" visual language. Prototyping across Python, Power BI, and Tableau, the team prioritized Box Plots to show percentile distributions and Scatter Plots to visualize feature relationships (e.g., Max Speed vs. Session Load). The plots below were built using a player-aggregated dataset, where each row contained the player’s metrics aggregated over all sessions.

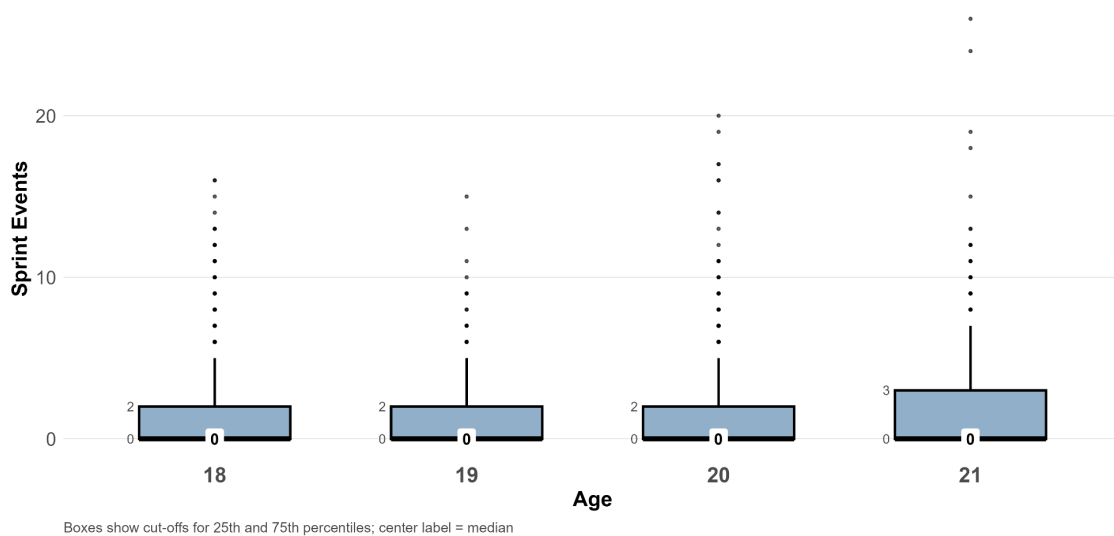
5.2 Deep Dive: Visual Insights & Interpretations



Average Top Speed by Age



Average Sprint Events by Age

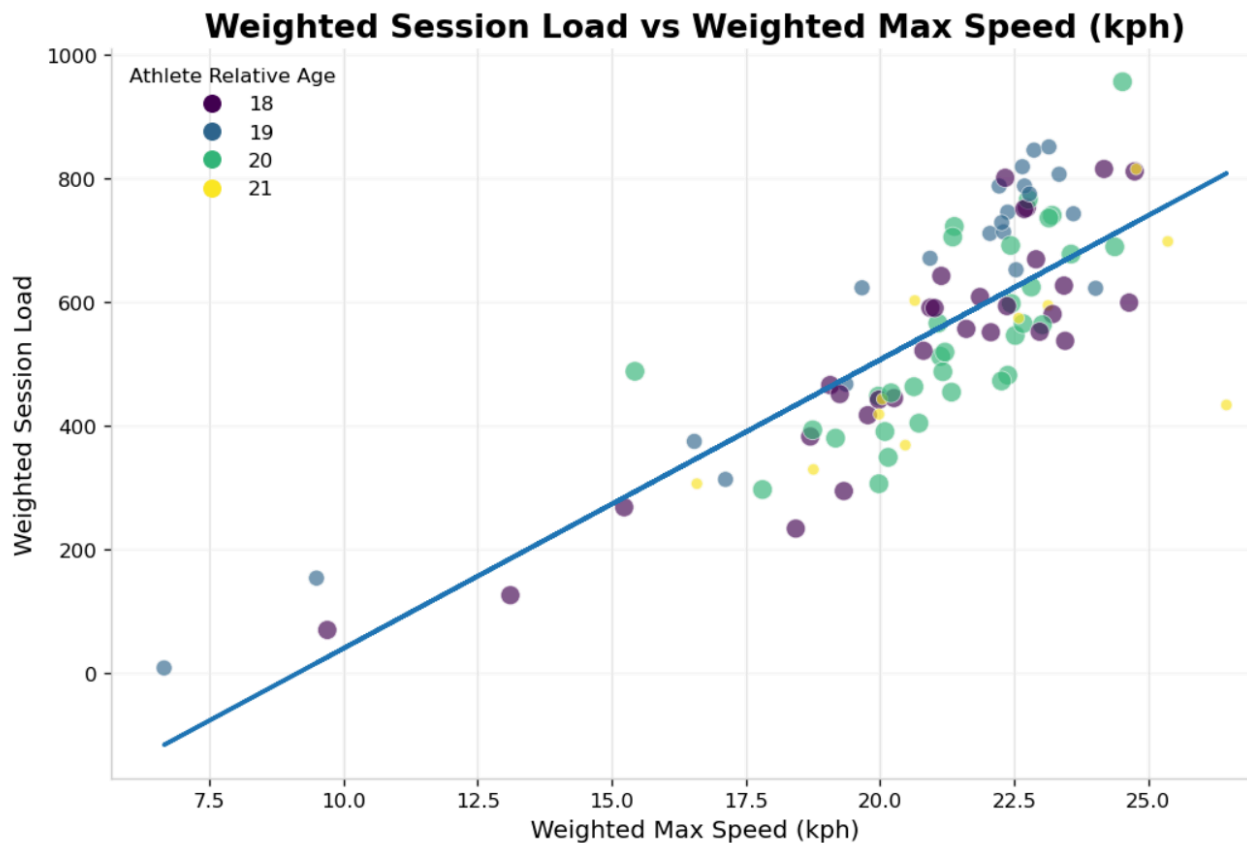


- The "Sophomore Peak" & "Senior Separation":** Boxplots for **Session Load** and **Distance** visually confirm the 19-year-old peak, where the median for 19-year-olds is about 35-40% higher than every other age group. The boxplots for **Top Speed** highlight that as athletes come into college at age 18 and as they end college at 21, the upper percentiles show a much greater "spread." This suggests that some freshmen come into college physically ready to compete, while others need additional time in training to reach their athletic peak. Additionally, top-tier seniors likely end up separating

themselves over time from the pack while other seniors may see declines due to injury or fatigue. Median **Top Speed** stays consistent at about 22 kph for each age group.

- **Sprint Events:** Increased upper percentile **Sprint Events** amongst the 21 year old age group suggests a similar adaptation to playstyle and game intensity noted in **Top Speed**. The increased range can also be attributed to increased desire or workload from an experienced player. That being said, most athletes (even seniors), record very few or no sprint events during their sessions which contributes to a median of zero for every age group.

- **Feature Correlation:**



The scatterplot reveals a nearly linear positive correlation between max speed and session load. This serves as a critical "Product Validation": PlayerData can confidently market **Session_load** as a reliable proxy for high-intensity sprinting.

5.3 Deployment Application

Coaches can use these visuals for **Training Optimization**. If an athlete is consistently in the 90th percentile for distance, the coach may reduce intensity to prevent injury. Conversely, those in the 10th percentile are identified as needing increased conditioning. This concept can be applied to the other metrics as well. Integration with the DE team's AI Agent would allow these visuals to be generated via plain-text questions, significantly improving in-app accessibility.

SECTION 6: LIMITATIONS & FUTURE RESEARCH

The project team has identified the following constraints and opportunities:

- **Positional & Session Context:** Without **Position Labels** (e.g., Goalkeeper vs. Midfielder) or **Session Type** (Game vs. Practice), benchmarks remain "noisy." Future work must include these tags to rank athletes fairly based on role-specific demands.
- **Temporal Depth:** The lack of longitudinal data prevents time-series modeling. Future research should track the current 18-year-old cohort over four years to see if the "Sophomore Peak" is a biological trend or a product of chance in this specific roster.
- **Multisport Expansion:** The current lack of data for Men's Soccer and other sports is a barrier. Future research must ingest wider datasets to generalize the benchmarking engine beyond collegiate women's soccer.
- **Data Lineage:** There is currently no "Data Versioning" or lineage tracking; future production-grade builds would require a clear audit trail of all upstream transformations applied by PlayerData before delivery.

APPENDIX

[Demo Application](#)

[Google Drive](#)

[Project GitHub](#)

[R Code for Metric Creation](#)